

Trabajo de Fin de Máster
Máster Universitario en Lógica, Computación e IA

A Study on Foundational Visual-Language
Models for Histopathology Image Analysis

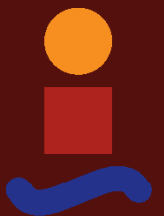
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Dpto. de Ciencias de la Computación e IA
Escuela Técnica Superior de Ingeniería Informática
Universidad de Sevilla

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Image Analysis

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*Leandro Candau Sánchez de Ybargüen
Sevilla, 2026*

Resumen

- Contexto: clasificación de imágenes histopatológicas con datos etiquetados limitados y etiquetas repetidas.
- Enfoque: aprendizaje contrastivo imagen-texto al estilo CLIP con codificadores fundacionales congelados.
- Datos: Kaggle LC25000, 5 clases de tejido de pulmón y colon.
- Diseño experimental: línea base controlada + matriz de ablaciones (una variable a la vez).
- Resultados (parciales): línea base 63.6 % accuracy / 62.7 % macro-F1; PLIP cero-disparo +7.3 macro-F1; ajuste fino parcial (últimas 30 capas) +27.1 macro-F1 (89.8 % / 89.8 %).

Abstract

- Context: histopathology image classification under limited labelled data and repeated labels.
- Approach: CLIP-style image-text contrastive learning with frozen foundation-model encoders.
- Data: Kaggle LC25000, five lung and colon tissue classes.
- Experimental design: a controlled baseline + a single-variable ablation matrix.
- Results (partial): baseline 63.6 % accuracy / 62.7 % macro-F1; PLIP zero-shot +7.3 macro-F1; partial fine-tuning (last 30 layers) +27.1 macro-F1 (89.8 % / 89.8 %).

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1 Introduction

Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision.

RADFORD ET AL., 2021

1.1 Motivation

- Clinical relevance of histopathology classification.
- Limited labelled data - recurring limitation in this domain.
- Foundation models - able to do different tasks with little task-specific supervision.
- Project question: "How does CLIP transfer to histopathology with different backbones? Where are its limits?"
- Framing: CLIP as the object of study, not the classifier. Findings should generalise to other scientific domains where short or missing labels are the norm.

1.2 Project Scope Evolution

- Initial direction -> semantic segmentation. Abandoned because annotated segmentation data was too scarce within the time budget.
- Pivot -> image-level classification, same domain, reframed question.
- Testbed: LC25000. Reference point: PLIP.

1.3 Research Questions

- RQ1: Frozen general-purpose CLIP transfer to LC25000 — quality and failure modes.
- RQ2: Pathology pretraining contribution (PLIP vs the general-purpose baseline).
- RQ3: Feature-adaptation contribution (partial unfreezing of the image encoder).
- RQ4: CLIP-paradigm contribution vs a plain image classifier on the same frozen features.
- RQ5: Stain normalization → shortcut-learning probe (Macenko / Reinhard / Vahadane head-to-head).
- RQ6: Biomedical text encoder (BioBERT / PubMedBERT) under short prompts.
- RQ7: Out-of-distribution transfer (external colon dataset).

1.4 Contributions

The central contribution of this project is a controlled measurement of how much each major design choice in a CLIP-style histopathology pipeline actually matters, with every claim framed as a delta over the same controlled baseline — frozen ResNet50 + frozen BERT, single fixed prompt, stratified 80/10/10 split on LC25000 — and iterated through one experiment in Chapter 4.

Measured claims:

- **Pathology-domain pretraining.** PLIP zero-shot vs the general-purpose baseline: +7.3 macro-F1 (69.8 % / 70.0 % vs 63.6 % / 62.7 %), with no LC25000 training. Gain concentrates on benign tissues; the two cancer subtypes (lung_aca, colon_aca) tie.
- **Feature-level adaptation.** Last 30 layers of ResNet50 unfrozen, LR dropped $10\times$ to 10^{-4} , everything else held constant: +27.1 macro-F1 (89.8 % / 89.8 %). Broad-based; every class $F_1 \geq 0.85$; lung_aca moves $0.48 \rightarrow 0.85$.
- **CLIP architecture vs plain CE classifier on the same features.** A 2×2 comparison at each adaptation budget. Frozen regime: tied within noise (+1.0 macro-F1 to plain CE, 0.6369 vs 0.6273). Unfreeze30 regime: CLIP wins +2.6 macro-F1 (0.8981 vs 0.8723), broad-based across classes. The CLIP architecture neither costs nor benefits accuracy at the frozen-features ceiling but gains a small consistent advantage when features adapt, plausibly via implicit regularisation from the InfoNCE loss.

Methodology:

- Single-variable-change protocol: each experiment differs from the baseline in exactly one design choice, so the resulting delta is attributable.
- Artifact bundle (fixed-seed split, weights, metric files, plots) versioned alongside the report in the project repository.
- Explicit written discussion of dataset and methodological limits constraining how far each delta can be generalised.

Pending ablations, scoped and implemented, to be folded in as measured claims once the corresponding run lands:

- Stain normalization head-to-head: Macenko, Reinhard, Vahadane vs raw inputs.
- Biomedical text encoder swap: BioBERT and PubMedBERT vs frozen BERT base.
- Out-of-distribution transfer of the baseline checkpoint to an external colon dataset.

1.5 Document Structure

- Chapter map to be added in the prose pass.

2 State of the Art

AI is undergoing a paradigm shift with the rise of models [...] that are trained on broad data at scale and are adaptable to a wide range of downstream tasks.

BOMMASANI ET AL., 2021

- Foundation models framing (Bommasani 2021).
- CLIP and multimodal contrastive learning (Radford 2021) — the paradigm under study.
- Self-supervised visual pretraining for medical imaging — DINO, DINOv2.
- Pathology shift: per-tile supervised CNNs → foundation models (2016–2024).
- Pathology-specific foundation models:
 - TransPath (2022) — transformer + SSL.
 - Phikon (2023) — masked image modelling.
 - PLIP (2023) — CLIP on medical Twitter.
 - CONCH (2024) — larger pathology VLM.
 - BiomedCLIP, MUSK — recent entrants.
- Open challenges: benchmarks, dataset bias, cross-institution generalization, compute scale.
- Where this project fits: small-scale empirical study of general-purpose foundation backbones on pathology classification, with single-variable ablations.

3 Methodology

3.1 Overview

- One-page chapter map + project question recap.
- Reading order: background → baseline → ablations.

3.2 Histopathology Digital Imaging

- H&E staining.
- Whole-Slide Imaging + patch extraction.
- Stain variability and normalization (Macenko, Reinhard, Vahadane).
- Why the domain is hard for general-purpose VLMs: domain gap from natural images, no per-image captions in clinical archives, dataset artifacts that can drive performance.

3.3 Contrastive Vision-Language Learning (CLIP)

3.3.1 Why CLIP?

- CLIP as object of study, not the classifier.
- Capabilities CLIP provides over a softmax head:
 - Open-vocabulary classification (new classes via prompts proven by adding new samples).
 - Semantic similarity output (cosine vs natural language).
 - Prompt engineering as a sensitivity lever.

3.3.2 Architecture and training signal

- Dual-encoder + shared embedding space.
- InfoNCE loss, learnable temperature.
- In-batch false-negative regime at small batch + few classes.
- Prompt-based zero-shot classification.
- CLIP-paper sample-level retrieval → doesn't apply (class-shared prompts).

3.4 Image Encoders

- ResNet (CNN baseline).
- Vision Transformer (ViT).
- DINO / DINOv2 (self-supervised).

Two independent encoders project image and text into a shared d -dimensional embedding space. The cosine-similarity matrix across a batch is the input to the symmetric InfoNCE loss.

Figure 3.1 CLIP dual-encoder architecture used in this project..

3.5 Text Encoders

- Transformer + BERT — text encoder used by baseline.
- BioBERT / PubMedBERT — biomedical text encoder swap ablation.

3.6 Reference baselines: PLIP and a plain image classifier

- PLIP zero-shot (Huang 2023) — pathology-pretrained CLIP, frozen, no LC25000 training.
- Plain image classifier — Dense(5)+softmax on frozen ResNet50 trained with cross-entropy; no text encoder, no CLIP architecture.
- Together: PLIP isolates the pretraining-domain contribution; the plain classifier isolates the CLIP-architecture contribution.

3.7 Dataset: Kaggle LC25000

- Borkowski 2019.
- 5 tissue classes (lung, colon).
- 25 000 patches from 1 250 originals via rotation/flip → no further geometric augmentation.
- Known limits: single institution, possible stain artifacts, no patient-level grouping.

Table 3.1 LC25000 class counts and stratified split sizes..

Class	Total	Train	Val	Test
benign lung tissue	5 000	4 000	500	500
lung adenocarcinoma	5 000	4 000	500	500
lung squamous cell carcinoma	5 000	4 000	500	500
benign colon tissue	5 000	4 000	500	500
colon adenocarcinoma	5 000	4 000	500	500
Total	25 000	20 000	2 500	2 500

Five-class sample: "benign lung", "lung adenocarcinoma", "lung squamous cell carcinoma", "benign colon", "colon adenocarcinoma".

Figure 3.2 Example image per class from LC25000..

3.8 Preprocessing and Input Pipeline

- Image: resize 224×224 , ResNet-style preprocessing, no further augmentation.
- Text: BERT WordPiece, fixed sequence length.
- Split: stratified 80/10/10, seed 42, persisted.

3.9 The Baseline

- Frozen ResNet50 + frozen BERT.
- Projection heads (Dense + LayerNorm) $\rightarrow$ 256-d shared space.
- Learnable temperature, CLIP-style init.
- Symmetric InfoNCE, AdamW, grad clipping.
- Single fixed prompt at train + eval.
- Float32 throughout.

Table 3.2 Baseline hyperparameters. Ablations change exactly one row..

Hyperparameter	Value
Image encoder	ResNet50 (ImageNet preset), frozen
Text encoder	BERT base uncased, frozen
Projection head	Dense(256, no bias) $\rightarrow$ LayerNorm
Embedding dimension	256
Image size	224×224
Text sequence length	24 WordPiece tokens
Batch size	32
Initial temperature	$\tau = 0.07$
Loss	symmetric InfoNCE
Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Gradient clipnorm	1.0
Max epochs	30
EarlyStopping patience	5
ReduceLROnPlateau	factor 0.5, patience 2, min 10^{-6}
Numerical precision	float32
Random seed	42

3.10 Evaluation Protocol

- Classification: `argmax cosine(image, class-prompt)`.
- Global: accuracy, balanced accuracy, macro-F1, weighted-F1.
- Per-class: precision, recall, F1.
- Confusion matrices: raw + row-normalized.
- Embeddings: shared image+prompt UMAP, centroid-to-prompt heatmap.
- Class-level retrieval, with explicit positive definition.

3.11 Experimental Design and Ablation Protocol

- Single-variable-change principle.
- Matrix grouped by axis of variation; full matrix in §4.1.2.
- Scope boundaries: no patient-level grouping, no human-expert evaluation, no architecture-controlled comparison vs CONCH / BiomedCLIP / MUSK. Deferred to future work.

3.12 Computational Setup and Reproducibility

- Hardware: Google Colab T4, local fallback.
- Stack: TensorFlow + KerasHub; PyTorch + transformers for PLIP only.
- Pinned library versions.
- Seeds + PYTHONHASHSEED + `tf.config.experimental.enable_op_determinism()`.

4 Results and Discussion

4.1 Experimental Overview

4.1.1 Research Questions

- RQ1: Frozen general-purpose CLIP transfer — quality + failure modes.
- RQ2: Pathology pretraining contribution (PLIP vs baseline).
- RQ3: Feature adaptation contribution (partial unfreezing).
- RQ4: CLIP paradigm value vs a plain image classifier on the same frozen features.
- RQ5: Stain normalization → stain shortcut + Macenko / Reinhard / Vahadane head-to-head.
- RQ6: Biomedical text encoder (BioBERT / PubMedBERT) under short prompts.
- RQ7: OOD transfer to external colon dataset.

4.1.2 Experiment Matrix

- Grouped by axis of variation. Baseline is the reference; others change exactly one variable.
- All rows on same 80/10/10 split unless testing external generalization.

4.2 Baseline: frozen ResNet50 + frozen BERT

4.2.1 Configuration

- Frozen ResNet50 (ImageNet) + frozen BERT base.
- Projection heads + 256-d shared space, learnable temperature.
- Symmetric InfoNCE, AdamW.
- Single fixed prompt; stratified 80/10/10 split (Table 3.1).

4.2.2 Training Behaviour

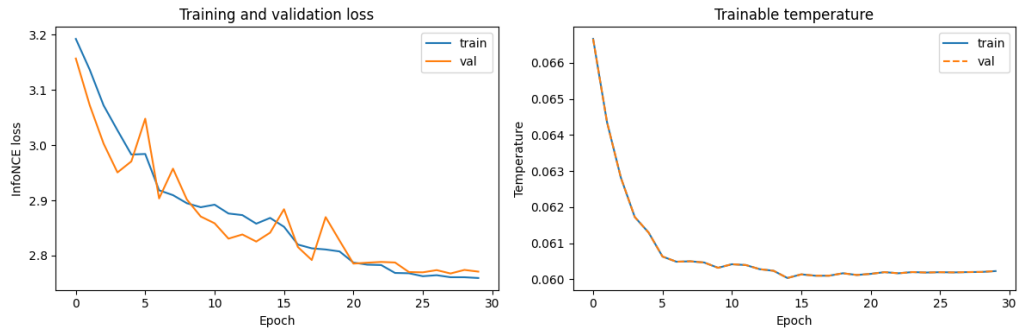
- Train/val loss + temperature trajectory (figure).

4.2.3 Classification Performance

- Global metrics (Table 4.2).
- Per-class P/R/F1 (Table 4.3).

Table 4.1 Experiment matrix grouped by axis of variation..

Experiment	Image enc.	Text enc.	Training	Axis tested
<i>Reference points (no LC25000 training)</i>				
PLIP zero-shot	ViT-B/16 (PLIP, frozen)	CLIP-text (PLIP, frozen)	none	pathology pretraining
OpenAI CLIP zero-shot	ViT-B/16 (frozen)	CLIP-text (frozen)	none	architecture-controlled
<i>Trained on LC25000</i>				
Baseline	ResNet50 frozen	BERT frozen	CLIP InfoNCE	reference
Plain image classifier	ResNet50 frozen	—	cross-entropy	value of CLIP paradigm
<i>Adaptation ablations</i>				
Partial fine-tuning	ResNet50 last-30 unfrozen	BERT frozen	CLIP InfoNCE	feature adaptation
Stain normalization	ResNet50 frozen	BERT frozen	CLIP InfoNCE	stain shortcut / method head-to-head
<i>Text-side ablation</i>				
Biomedical text encoder	ResNet50 frozen	BioBERT / PubMedBERT frozen	CLIP InfoNCE	biomedical text pretraining
<i>Generalization</i>				
External colon dataset	ResNet50 frozen	BERT frozen	baseline checkpoint	OOD transfer
<i>Image encoder family</i>				
DINOv2 image encoder	DINOv2 frozen	BERT frozen	CLIP InfoNCE	self-supervised pretraining

**Figure 4.1** Baseline training and validation InfoNCE loss (left) and learnable temperature τ (right) against epoch. τ drifts from the CLIP initialisation of 0.07 toward its converged value..**Table 4.2** Baseline global classification metrics on the test set..

Metric	Value
Accuracy	0.6356
Balanced accuracy	0.6356
Macro F1	0.6273
Weighted F1	0.6273

4.2.4 Confusion Matrices

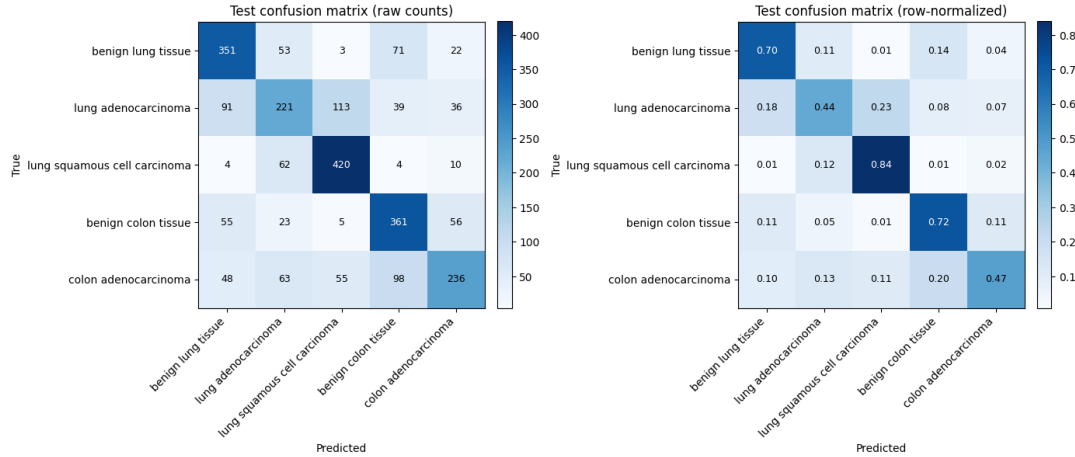
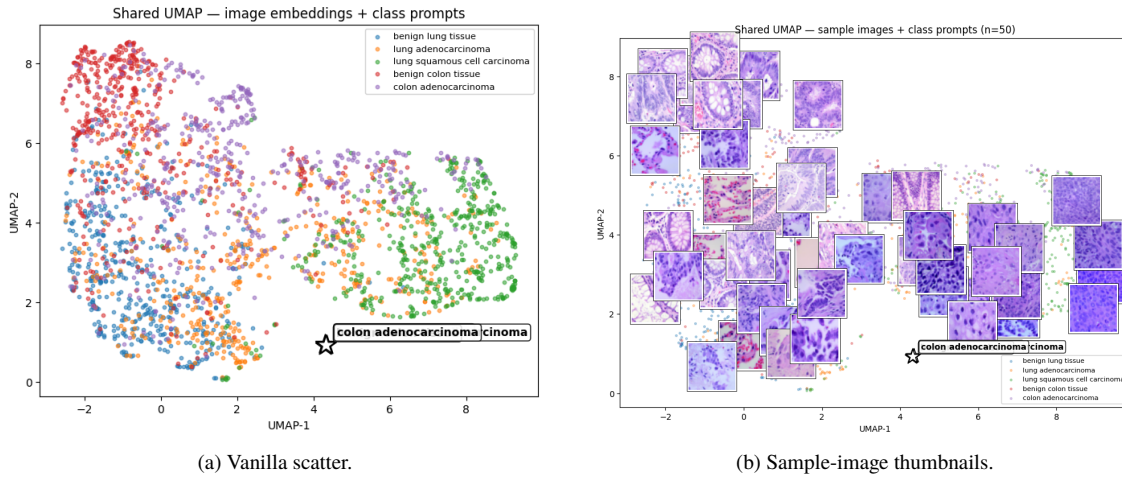
- Raw + row-normalized (figure pair).

4.2.5 Embedding Visualizations

- Shared image+prompt UMAP (two views: vanilla scatter and sample-image thumbnail overlay).
- Centroid-to-prompt similarity heatmap (figure).

Table 4.3 Baseline per-class classification metrics..

Class	Precision	Recall	F1	Support
benign lung tissue	0.639	0.702	0.669	500
lung adenocarcinoma	0.524	0.442	0.479	500
lung squamous cell carcinoma	0.705	0.840	0.766	500
benign colon tissue	0.630	0.722	0.673	500
colon adenocarcinoma	0.656	0.472	0.549	500
Macro average	0.631	0.636	0.627	2 500

**Figure 4.2** Baseline confusion matrices. Left: raw counts. Right: row-normalised by true class..**Figure 4.3** Shared UMAP of test image embeddings and the five class-prompt embeddings (cosine metric), in two views from the same fit. (a) Per-class image scatter with class-prompt embeddings marked. (b) Same coordinate system overlaid with ~ 50 sample image thumbnails (no two within a minimum distance) and prompt markers on top — adapted from the reference notebook supplied by the tutor..

4.2.6 Class-Level Retrieval

- Positives = class-shared (not per-sample). Not CLIP-paper retrieval.
- $R@K + P@K$ (table).

4.2.7 Baseline Interpretation

- Class-by-class confusion structure: lung_aca worst, lung_scc best.

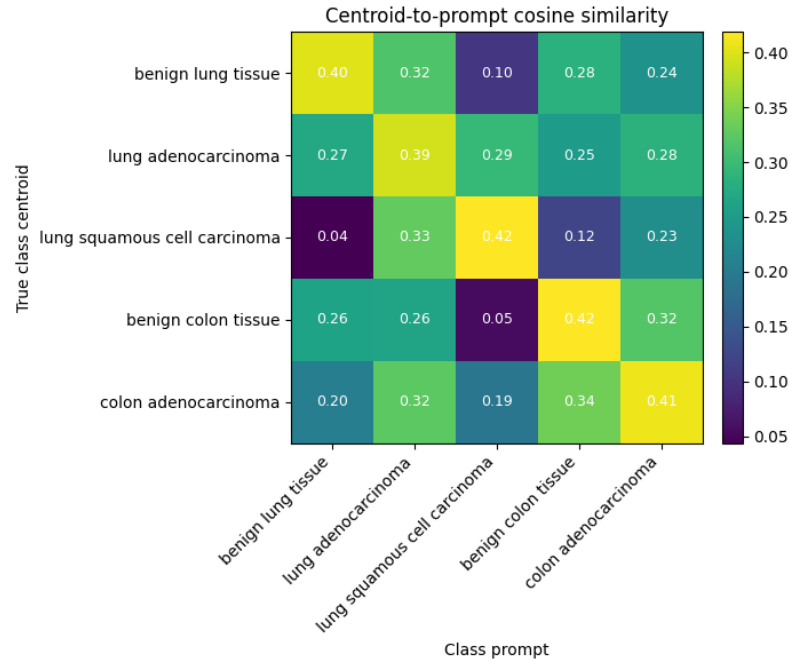


Figure 4.4 Baseline centroid-to-prompt cosine similarity heatmap. Rows: image-embedding centroid for each true class. Columns: the five class-prompt embeddings. Cell values: cosine similarity. A diagonal-heavy matrix indicates that each class’s images align most strongly with their own prompt..

Table 4.4 Baseline class-level retrieval..

Metric	K=1	K=5	K=10	K=50
Image → Class prompt R@K	0.6356	1.0000	—	—
Class prompt → Image P@K	0.8000	0.7200	0.8000	0.8400

- Stain artifacts → how much they plausibly contribute.
- Embedding structure → what it says about ImageNet+BERT → pathology transfer.

4.3 PLIP zero-shot comparison

4.3.1 Setup

- PLIP (Huang 2023): CLIP on ~200k pathology Twitter pairs.
- Loaded `vinid/plip`, evaluated zero-shot on same test split + prompts.
- Isolates pathology-pretraining contribution.

4.3.2 Headline numbers vs baseline

Table 4.5 Pathology pretraining contribution to frozen CLIP transfer..

Metric	Baseline (general)	PLIP (pathology)	Δ
Accuracy	0.6356	0.6980	+0.062
Balanced accuracy	0.6356	0.6980	+0.062
Macro F1	0.6273	0.7004	+0.073
Weighted F1	0.6273	0.7004	+0.073

4.3.3 Per-class

- Big gains on benign tissues (lung F1 0.67→0.87; colon F1 0.67→0.80).
- Cancer subtypes essentially tied (lung_aca F1 0.48→0.48; colon_aca F1 0.55→0.56).
- Lung SCC recall jumps (0.84→0.93), precision drops (0.70→0.68) → overcalls SCC.

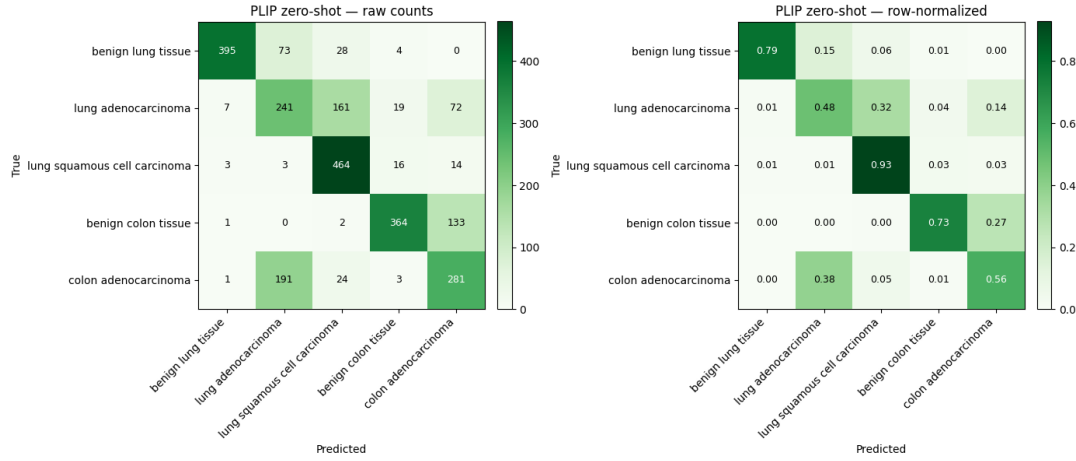


Figure 4.5 PLIP zero-shot confusion matrices, same five classes as the baseline. Left: raw counts. Right: row-normalised by true class..

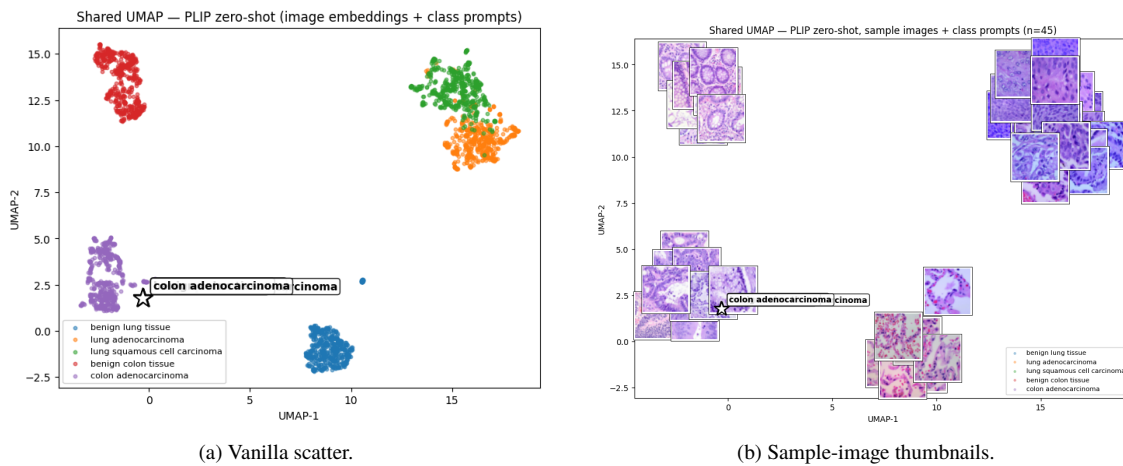


Figure 4.6 PLIP zero-shot shared UMAP of test image embeddings and the five class-prompt embeddings (cosine metric). Image clusters separate by class meaningfully better than the baseline (Figure 4.3), but the five prompt markers still collapse into a near-singular point — the same short-prompt + frozen text encoder degeneracy as the baseline. PLIP earns its +7.3 macro-F1 by producing better image embeddings, not by separating prompts..

4.3.4 Retrieval contrast

- I→T R@1 = 0.6980 (matches accuracy).
- T→I P@K *worse* than the baseline (P@1: 0.40 vs 0.80; P@50: 0.53 vs 0.84).
- Reading: the baseline was trained on LC25000 → embedding space tuned for in-distribution ranking. PLIP zero-shot → wins classification, loses fine ranking.

4.3.5 Interpretation

- Pathology pretraining ≈ +7.3 macro-F1 on frozen CLIP transfer.
- Gain in benign-vs-malignant, not in subtype discrimination.

- Image clusters in the shared UMAP separate visibly better than the baseline (Figure 4.6), but the prompt markers still collapse — domain pretraining helps the image side, not the (still-degenerate) text side.
- Feature adaptation confirmed as the bigger lever in §4.5; PLIP’s frozen-domain gain is one-third of unfreezing’s frozen-vs-adapted gain.

4.4 Plain image classifier on the same features

- Drops the CLIP architecture: no text encoder, no projection heads, no learnable temperature, no contrastive loss.
- Replaces with: ResNet50 $\rightarrow$ GAP $\rightarrow$ BatchNorm $\rightarrow$ Dense(5) trained with sparse cross-entropy.
- Run at both feature-adaptation budgets (frozen and unfreeze30) so the comparison vs the CLIP variants in §4.2 and §4.5 is a complete 2×2 .
- Isolates what the CLIP architecture itself contributes once the image encoder, data and adaptation budget are held constant.

Table 4.6 CLIP vs plain CE classifier on the same ResNet50 backbone at each feature-adaptation budget. Frozen regime: tied within noise. Unfreeze30 regime: CLIP wins +2.6 macro-F1..

	Frozen		Unfreeze last 30	
Architecture	Acc.	Macro F1	Acc.	Macro F1
CLIP (InfoNCE)	0.6356	0.6273	0.8984	0.8981
Plain (CE softmax)	0.6440	0.6369	0.8728	0.8723
Δ (CLIP – plain CE)	−0.0084	−0.0096	+0.0256	+0.0258

4.4.1 Interpretation

- Frozen regime: CLIP and plain CE tie within noise. Both saturate at the ImageNet-features ceiling of ~ 0.63 ; the architecture choice is invisible when the feature space is fixed.
- Unfreeze30 regime: CLIP outperforms plain CE by +2.6 macro-F1, broad-based across every class (benign_lung +0.018; lung_aca +0.022; lung_scc +0.020; benign_colon +0.038; colon_aca +0.031).
- Plausible mechanism: InfoNCE with five shared class-prompt targets acts as implicit regularisation against per-image memorisation. With ~ 14 M trainable parameters in the unfrozen tail, requiring every sample of a class to align with the same prompt limits how much sample-specific information the model can store. The benefit only materialises once the model has the capacity to overfit, which is why the frozen regime sees no difference.
- The CLIP paradigm’s value on a closed-set 5-class task is not raw accuracy. The +2.6 macro-F1 bonus under feature adaptation is real but secondary — the paradigm’s load-bearing contribution is capabilities a plain classifier cannot provide: open-vocabulary classification at inference, image–text retrieval, prompt-tuning as a controlled lever.

4.5 Feature-level adaptation: partial unfreezing

- Same arch/loss/prompts/data/split as the baseline; only the trainable mask on ResNet50 changes.
- Last 30 of 178 ResNet50 layers unfrozen (≈ 14.4 M extra trainable params on top of the projection heads).
- BatchNorm in the unfrozen tail updates its running stats during training (training mode) and uses them at eval.
- LR dropped $10\times$ to 10^{-4} to protect the pretrained features.
- +27.1 macro-F1 over the baseline ($\approx 3.7\times$ the PLIP zero-shot gain).

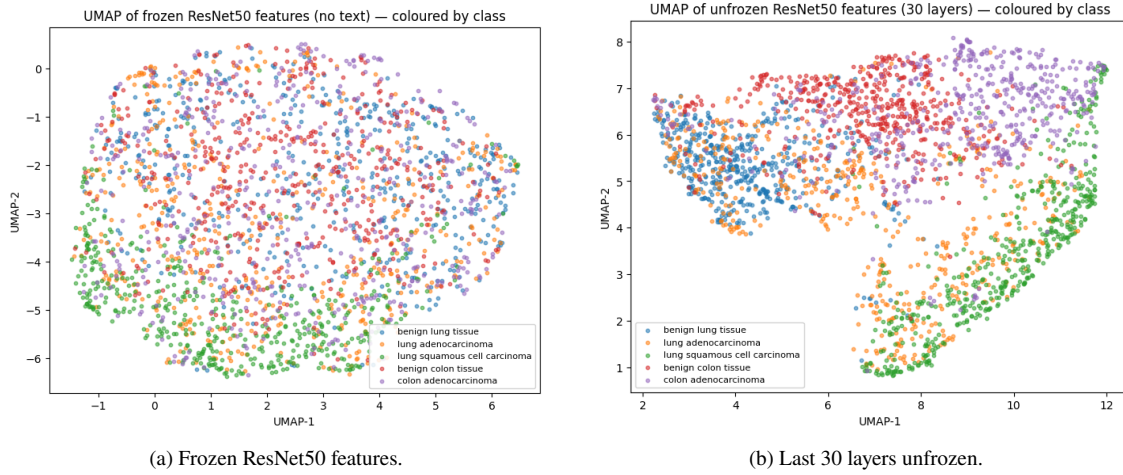


Figure 4.7 Plain-classifier UMAP of test image features, coloured by true class. No text encoder in this ablation, so no prompt markers — this is an image-side view only. (a) Frozen features show weak, tangled class structure consistent with the ~ 0.64 classification ceiling. (b) With the last 30 layers unfrozen, classes resolve into five well-separated clusters, consistent with the ~ 0.87 accuracy..

Table 4.7 Partial fine-tuning vs frozen baseline: global classification metrics on the held-out test set..

Metric	Baseline	Unfreeze last 30
Accuracy	0.6356	0.8984
Balanced accuracy	0.6356	0.8984
Macro F1	0.6273	0.8981
Weighted F1	0.6273	0.8981

- Broad-based across classes: every class $F_1 \geq 0.85$.
- Lung_aca: F_1 0.48 $\rightarrow$ 0.85 (worst baseline class; still hardest but no longer a failure mode).
- +0.25 macro-F1 over the frozen-features ceiling measured by the plain CE classifier (~ 0.64 , see §4.4). Headroom to fully-fine-tuned literature results on LC25000 (~ 0.95 – 0.99 , end-to-end training, often with stronger backbones) is harder to pin down precisely without running that regime ourselves.

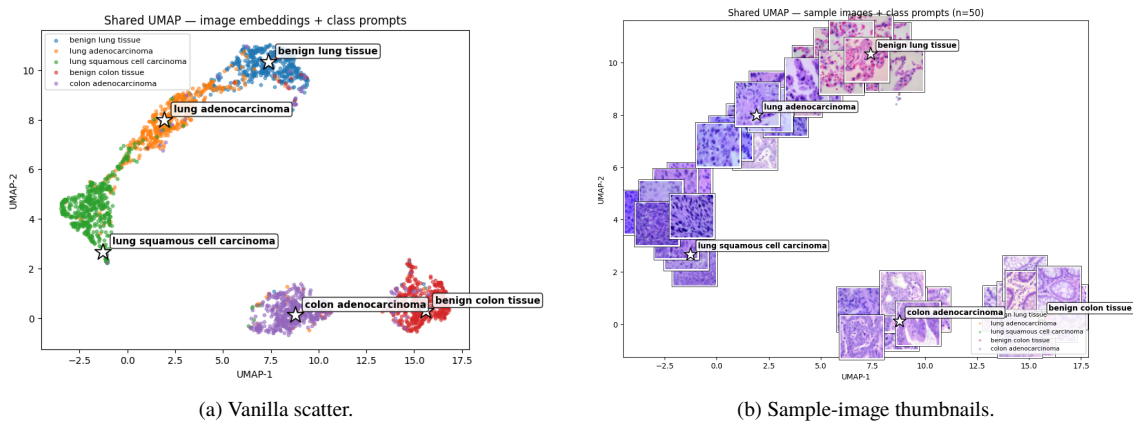


Figure 4.8 Unfreeze30 shared UMAP of test image embeddings and the five class-prompt embeddings (cosine metric), in two views from the same fit. Compare with the baseline UMAP in Figure 4.3 — the unfrozen-tail embedding space shows tighter, better-separated class clusters and prompt markers that sit clearly inside their own cluster..

4.6 Stain normalization ablation

4.6.1 Setup

- Same arch/loss/prompts/split/optimizer as the baseline (§4.2); only the image inputs change.
- Head-to-head between Macenko [1], Reinhard, and Vahadane on the *same* fixed reference patch (colon_n/colonnn1.jpeg, resized to 224×224).
- Each method is fit once on the reference, then applied deterministically to every train/val/test image. Normalization is precomputed and cached on Drive so re-runs of figures or downstream evals are cheap.
- The frozen-encoder regime is held fixed so the delta isolates the stain-normalization method itself, not feature adaptation.

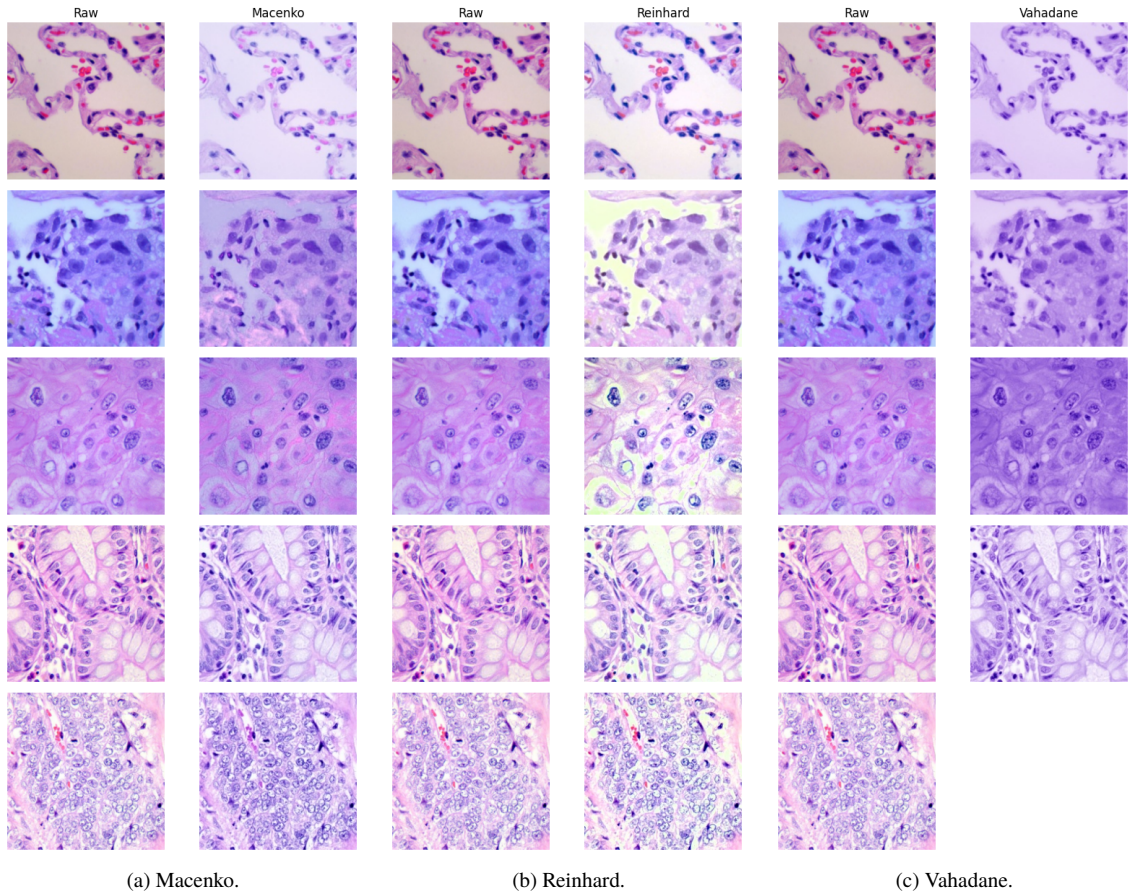


Figure 4.9 Before (left column of each pair) / after (right column) normalization, one example per class. Macenko (SVD on stain vectors in OD space) and Vahadane (sparse NMF on the same OD matrix) both decompose into haematoxylin and eosin channels and re-render with the reference’s basis; Reinhard matches first- and second-order LAB-space statistics. The bottom-right Vahadane panel (colon_aca) shows a failed re-render, discussed in the interpretation..

4.6.2 Headline numbers vs baseline

4.6.3 Per-class

The three methods give visibly different per-class signatures.

Table 4.8 Three-way stain-normalization head-to-head on the same frozen CLIP+ResNet50+BERT pipeline. Macenko delivers $\sim 3\times$ Vahadane’s gain on macro-F1, and $\sim 4.7\times$ Reinhard’s..

Metric	Baseline	Macenko	Δ_M	Reinhard	Δ_R	Vahadane	Δ_V
Accuracy	0.6356	0.7384	+0.103	0.6564	+0.021	0.6684	+0.033
Balanced acc.	0.6356	0.7384	+0.103	0.6564	+0.021	0.6684	+0.033
Macro F1	0.6273	0.7390	+0.117	0.6526	+0.025	0.6644	+0.037
Weighted F1	0.6273	0.7390	+0.117	0.6526	+0.025	0.6644	+0.037

Table 4.9 Per-class F1 under each normalization. Macenko gains across four of five classes (loses 3.8 pts on lung_scc); Reinhard’s gain is colon-only and it actively *loses* 6.1 pts on benign lung; Vahadane is mixed across classes, with a colon_aca gain comparable to Reinhard (+0.15) but essentially flat on the lung classes..

Class	Baseline F1	Macenko F1	Δ_M	Reinhard F1	Δ_R	Vahadane F1	Δ_V
benign lung tissue	0.669	0.885	+0.216	0.608	-0.061	0.682	+0.013
lung adenocarcinoma	0.479	0.575	+0.096	0.482	+0.003	0.464	-0.015
lung squamous cell carcinoma	0.766	0.728	-0.038	0.756	-0.010	0.788	+0.022
benign colon tissue	0.673	0.775	+0.102	0.727	+0.054	0.689	+0.016
colon adenocarcinoma	0.549	0.732	+0.183	0.690	+0.141	0.698	+0.149

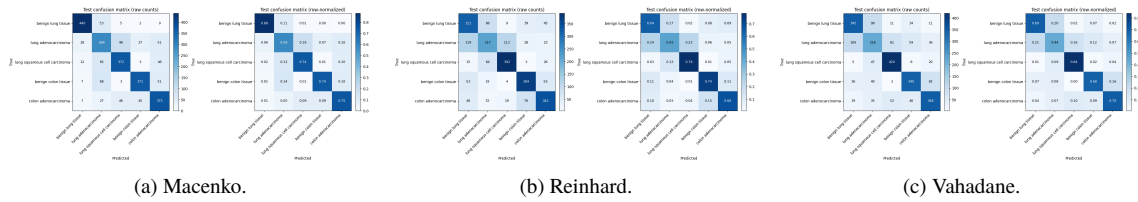


Figure 4.10 Confusion matrices (row-normalised) under each stain normalization, same five classes as the baseline..

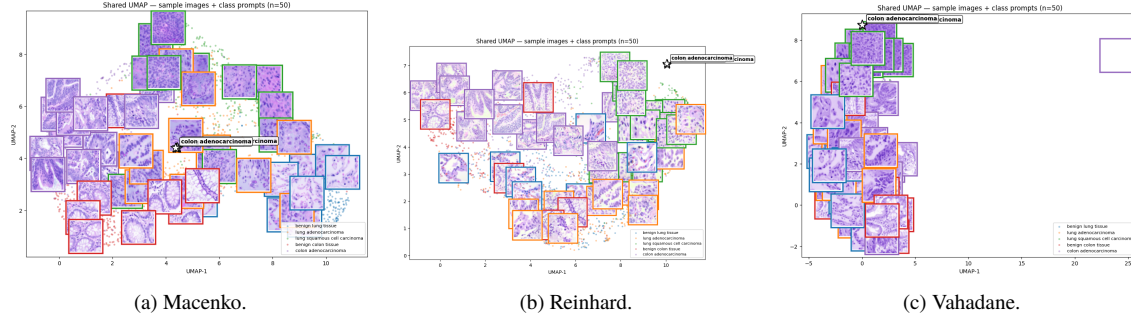


Figure 4.11 Shared UMAP (image embeddings + class-prompt markers, cosine metric) under each normalization. Compared to the baseline UMAP (Figure 4.3), Macenko shows clearly tighter benign-vs-malignant separation; Reinhard and Vahadane improve over baseline more modestly. Prompt markers still collapse in all three cases — every gain is image-side..

4.6.4 Retrieval contrast

- I $\rightarrow$ T R@1 mirrors classification accuracy by construction (Macenko 0.738, Vahadane 0.668, Reinhard 0.656); R@5 = 1.0 in all three cases with five prompts.
- T $\rightarrow$ I class-level retrieval is stronger than baseline for all three methods: Macenko P@1/5/10/50 = 1.00/0.92/0.88/0.90; Vahadane 1.00/0.92/0.90/0.86; Reinhard 1.00/0.96/0.92/0.85; baseline 0.80/0.72/0.80/0.84. Even Reinhard, whose classification gain is small, sharpens within-class ranking.

4.6.5 Composition with feature adaptation (best case)

The 3-way head-to-head above isolates the stain-normalization method at a fixed (frozen) feature-adaptation budget. We now combine the best stain method (Macenko) with the best feature-adaptation regime (unfreeze30,

§4.5) into a single run — Macenko-normalised inputs feeding a CLIP model whose last 30 ResNet50 layers are trainable, everything else identical to the baseline. This is the load-bearing checkpoint for the OOD evaluation in §4.8.

Table 4.10 Composition of the two strongest single levers. Combining Macenko (best stain method) with partial fine-tuning gives the highest macro-F1 measured in this study and is the headline RQ5 number..

Variant	Accuracy	Macro F1	Δ vs baseline
Baseline	0.6356	0.6273	—
Macenko (frozen)	0.7384	0.7390	+0.117
Unfreeze30 (raw inputs)	0.8984	0.8981	+0.271
Macenko + Unfreeze30	0.9180	0.9180	+0.291

Table 4.11 Macenko + Unfreeze30 per-class metrics. The lung_aca class — the worst across every other variant — is the only one that doesn’t clear $F1 = 0.85$, but it still gains +0.37 over the baseline..

Class	Precision	Recall	F1	Δ vs baseline
benign lung tissue	0.971	0.996	0.983	+0.314
lung adenocarcinoma	0.853	0.838	0.846	+0.366
lung squamous cell carcinoma	0.856	0.902	0.878	+0.112
benign colon tissue	0.953	0.940	0.947	+0.274
colon adenocarcinoma	0.960	0.914	0.936	+0.388

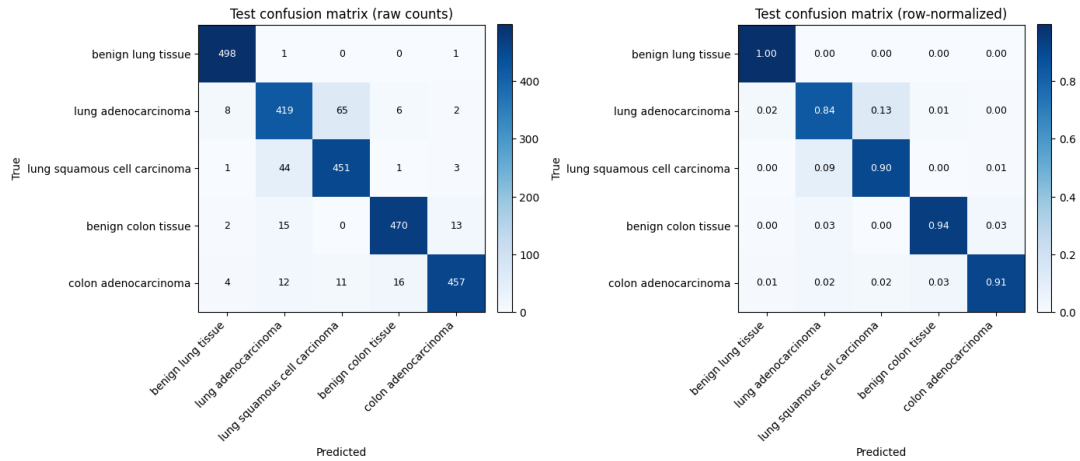


Figure 4.12 Confusion matrices for Macenko + Unfreeze30. Off-diagonal mass concentrates on lung_aca vs lung_scc, mirroring the residual confusion in the unfreeze30-only run..

Reading. Macenko (+0.117) and unfreeze30 (+0.271) compose with diminishing returns: combining them gives +0.020 over unfreeze30 alone and +0.179 over Macenko alone. Feature adaptation does the bulk of the work — once the backbone can shift its features to the LC25000 distribution, the input-side shift from stain normalization adds only a few points on top. But those few points are clean and broad-based (gains on 4 of 5 classes), so the composition is unambiguously preferred over either lever in isolation. This is the variant used as the LC25000-trained checkpoint in the OOD transfer evaluation.

4.6.6 Interpretation

- Macenko clears the frozen-features ceiling by +11.7 macro-F1 — larger than the PLIP zero-shot gain (+7.3, §4.3) and obtained without changing the backbone or any pretraining.
- Vahadane and Reinhard add only +3.7 and +2.5 macro-F1. The $3\times$ gap to Macenko is the informative comparison: Macenko and Vahadane share the same principle (decompose each image’s optical-density matrix into a H/E stain matrix and a per-pixel concentration matrix, re-render with the reference’s basis) and differ only in solver — Macenko uses closed-form SVD, Vahadane uses iterative sparse NMF. On LC25000, the NMF solver fails to converge on a fraction of patches and returns near-zero

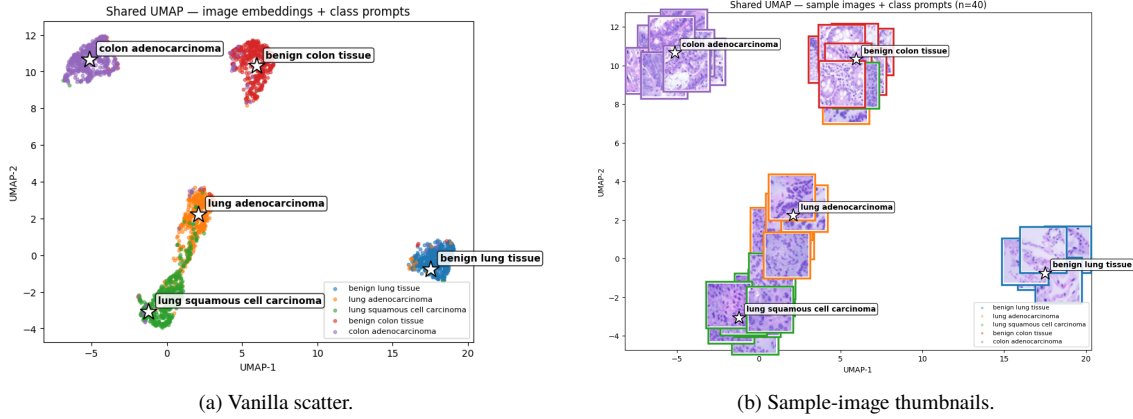


Figure 4.13 Macenko + Unfreeze30 shared UMAP. Compared to the unfreeze30-only UMAP (Figure 4.8) the class clusters are slightly tighter and the lung_aca / lung_scc boundary is the only remaining noisy region. Class-prompt markers sit clearly inside their own clusters..

concentrations, visible directly in Figure 4.9(c) bottom-right as a near-blank re-render. The realised stain-normalization gain therefore depends on solver stability, not just on the underlying principle.

- RQ5 reading: the large method-specific gain on frozen general-purpose features is consistent with the model relying on inter-image colour variation as a partial shortcut. Macenko removes that signal cleanly; Reinhard removes it together with some legitimate diagnostic colour cues (benign lung F1 drops 6.1 points under Reinhard); Vahadane removes it on most patches but corrupts a fraction. The shortcut hypothesis survives across all three.
- Per-class signatures point at the reference: the reference patch is a benign colon image, and the colon classes gain under every method (Macenko +0.183, Reinhard +0.141, Vahadane +0.149 on colon_aca). Lung classes only gain meaningfully under Macenko, which keeps tissue types in roughly the right corner of feature space; under Reinhard and Vahadane they are flat or slightly degraded.
- Read with PLIP: pathology pretraining shifts the image-feature distribution toward H&E content; stain normalization shifts the input distribution toward a fixed H&E reference. Both gains accrue on the image side of the UMAP, not the text side — the prompt-marker collapse persists under every normalisation tested.
- Caveats: single seed per condition, single reference patch. Reference rotation and a more robust NMF setup (higher `max_iter`, different init) are left as future work.

4.7 Biomedical text encoder ablation

- Same arch/loss/prompts/split as the baseline; only the text encoder changes.
- Swap BERT base → BioBERT, then → PubMedBERT (frozen in both cases).
- Tests whether biomedical text pretraining helps under short class-name prompts.
- Result: *TBD*.

4.8 Out-of-distribution generalization

4.8.1 Setup

- External dataset: **NCT-CRC-HE-7K** (Kather et al. 2019, Zenodo record 1214456), the standardised held-out test split of the Heidelberg colorectal histopathology dataset. 7 180 patches across 9 tissue classes, acquired and stained at a different institution than LC25000.
- Evaluation restricted to NCT classes NORM (normal colon mucosa) and TUM (colorectal adenocarcinoma epithelium), which map directly to LC25000's `benign colon tissue` and `colon adenocarcinoma`. Total OOD test set: $741 + 1\,233 = 1\,974$ patches.

- All five LC25000 class prompts are kept at evaluation, so each variant is evaluated under the same prompt regime it was trained with. A misclassification into a lung class counts as wrong.
- Pure inference: every variant reuses its checkpoint from the in-distribution experiments above. No fine-tuning on NCT.

4.8.2 Variants compared

We evaluate four LC25000-trained CLIP variants, a domain-pretrained reference (PLIP zero-shot), and — to disambiguate the role of OOD-time Macenko normalization — two extra ablation rows where the Macenko-trained checkpoints are fed the *raw* NCT inputs instead of NCT-normalised-with-LC25000’s Macenko fitter:

Table 4.12 OOD transfer to NCT-CRC-HE-7K (NORM + TUM, 1 974 patches). PLIP zero-shot wins across the board; the best LC25000-trained variant is the raw-trained Unfreeze30 (frozen-Macenko-trained variants fail catastrophically with their training-time normalization at OOD)..

Variant	OOD inputs	Accuracy	Balanced acc.	Macro F1
Baseline CLIP (raw, frozen)	raw	0.1327	0.1399	0.2134
Macenko CLIP (frozen)	Macenko	0.0000	0.0000	0.0000
Unfreeze30 CLIP (raw)	raw	0.3865	0.3167	0.3309
Macenko + Unfreeze30 (best in-distribution)	Macenko	0.0000	0.0000	0.0000
<i>Disambiguation: Macenko-trained checkpoints fed raw NCT inputs</i>				
Macenko ckpt, raw NCT inputs	raw	0.0922	0.0988	0.1538
Combo ckpt, raw NCT inputs	raw	0.1449	0.1216	0.2011
<i>Domain-pretrained reference</i>				
PLIP zero-shot	raw (PLIP processor)	0.5572	0.6095	0.6610

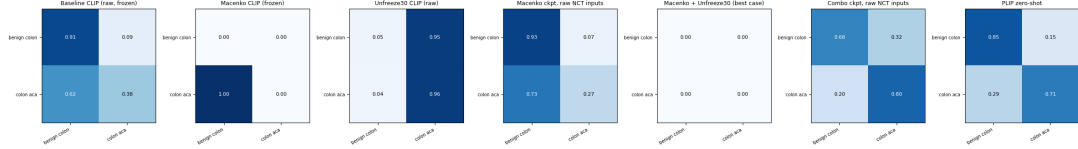


Figure 4.14 Row-normalised confusion matrices on NCT-CRC-HE-7K (NORM + TUM subset). Cells in the two displayed columns sum to less than 1 when the model predicts a non-colon LC25000 class (lung_aca, lung_scc, benign_lung); that missing probability mass is the rate at which the model leaves the colon label space entirely. The two Macenko-trained variants with Macenko-normalised inputs predict outside the colon classes on essentially every patch..

4.8.3 Interpretation

- **Organ-level transfer works; within-organ transfer does not.** A close reading of the cross-distribution UMAP (Figure 4.16) and the confusion matrices reveals that the LC25000-trained variants do not fail uniformly. The non-broken variants (baseline, Unfreeze30, the two raw-NCT disambiguation rows) all separate LC25000 lung from LC25000 colon in their embedding space, and they all place the NCT colon patches on the colon side — the model has carried over the organ distinction. What it has not carried over is the within-colon distinction: the confusion matrices show baseline predicting nearly everything as *benign_colon* regardless of true label, and Unfreeze30 predicting nearly everything as *colon_aca*. The errors are *within* colon, not into lung. This refines the headline finding: LC25000 training transfers organ-level features but does not transfer the cellular cues that separate normal mucosa from adenocarcinoma epithelium under NCT’s tighter label semantics (caveat 3).
- **PLIP wins by a wide margin.** A domain-pretrained foundation model with no LC25000 fine-tuning (0.661 macro-F1) beats every LC25000-trained variant on this OOD test, including the best in-distribution model (0.918 in-distribution, 0.000 OOD). The PLIP UMAP (Figure 4.15) and cross-distribution UMAP (Figure 4.16) are the only ones with both clean organ separation *and* within-organ NORM/TUM separation. Pathology pretraining is the only ingredient that produces representations which transfer at both levels.

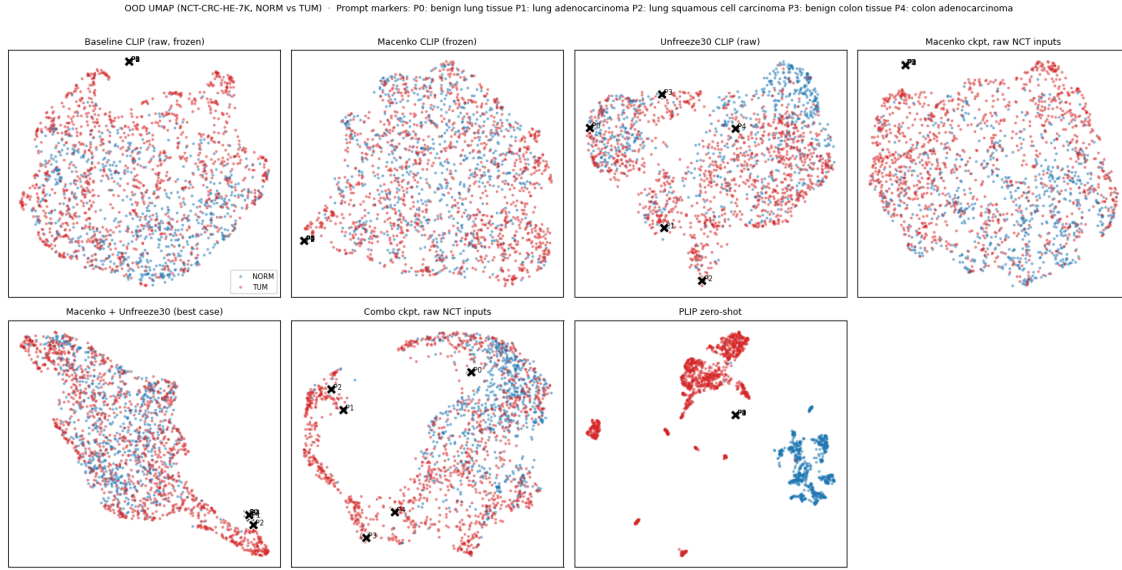


Figure 4.15 Per-variant UMAP of OOD image embeddings (shared fit with the five LC25000 class-prompt markers $P_0 \dots P_4$, cosine metric). Image points are coloured by NCT class (NORM / TUM). The six LC25000-trained variants all show a single dense interleaved blob — the model’s representation cannot distinguish NCT NORM from NCT TUM regardless of training recipe. Only PLIP (bottom row, third panel) produces two visibly separated colon clusters with the prompt markers anchored between them..

- **The OOD-time Macenko transform is itself catastrophic.** Both Macenko-trained checkpoints drop to 0% macro-F1 when their training-time Macenko normalization is also applied at OOD. The most likely mechanism is double-normalization: NCT-CRC-HE was already colour-normalised against its own internal reference in the Kather 2019 release, and applying our LC25000-fit Macenko on top shifts NCT pixels into a region of input space the model has never seen, despite the visual output of the second normalization step looking superficially reasonable (Figure 4.9 analogue not shown for NCT).
- **Removing Macenko at OOD only partially recovers the Macenko checkpoints.** The two disambiguation rows (Macenko ckpts fed raw NCT) recover from 0% to 9–14% macro-F1, but they remain below the raw-trained Unfreeze30 variant (0.331) by a factor of two. The Macenko-trained *features* themselves are more specialised to the LC25000-Macenko’s input distribution than raw-trained features are to raw LC25000.
- **A trade-off, not a free lunch.** In-distribution, Macenko + Unfreeze30 was the highest-performing model in the entire study (0.918 macro-F1, +0.291 over baseline, §4.6.5); OOD, the same model is among the worst. The training recipe that maximises in-distribution accuracy also maximises domain specialization. This is the central finding of RQ7: *stain normalization is an in-distribution lever; under domain shift, it can become a liability*.
- **Caveats.** (1) We restrict the eval to NORM + TUM because no other NCT class maps to an LC25000 prompt; a richer label space (e.g., adding “stroma” / “lymphocyte” prompts to the model at training time) might widen the comparison. (2) NCT-CRC-HE-7K is itself colour-normalised; an experiment on the unnormalised NCT-CRC-HE-100K-NONORM variant would test whether the Macenko-at-OOD breakage is specific to double-normalization. (3) The class labels themselves are not fully aligned across datasets: NCT TUM is specifically “colorectal adenocarcinoma epithelium” (with separate parallel classes for cancer-associated stroma, mucus, debris, lymphocytes), whereas LC25000 colon adenocarcinoma is a patient-level patch label that may include adjacent stroma, lymphocytes, and mucus together. Same for NORM vs benign colon tissue. Part of the transfer gap is a label-granularity mismatch, not just imaging shift. (4) Magnification is explicit for NCT ($0.5 \mu\text{m}/\text{px}$, $\approx 20\times$) but not stated in the LC25000 paper; visual inspection suggests $\sim 20\times$ but a small mismatch in microns-per-pixel would shift the effective field of view at the model’s input. (5) Single seed per condition; the disambiguation rows share a checkpoint with their main counterparts, so the comparison there is not an independent replicate but a controlled input-side ablation.

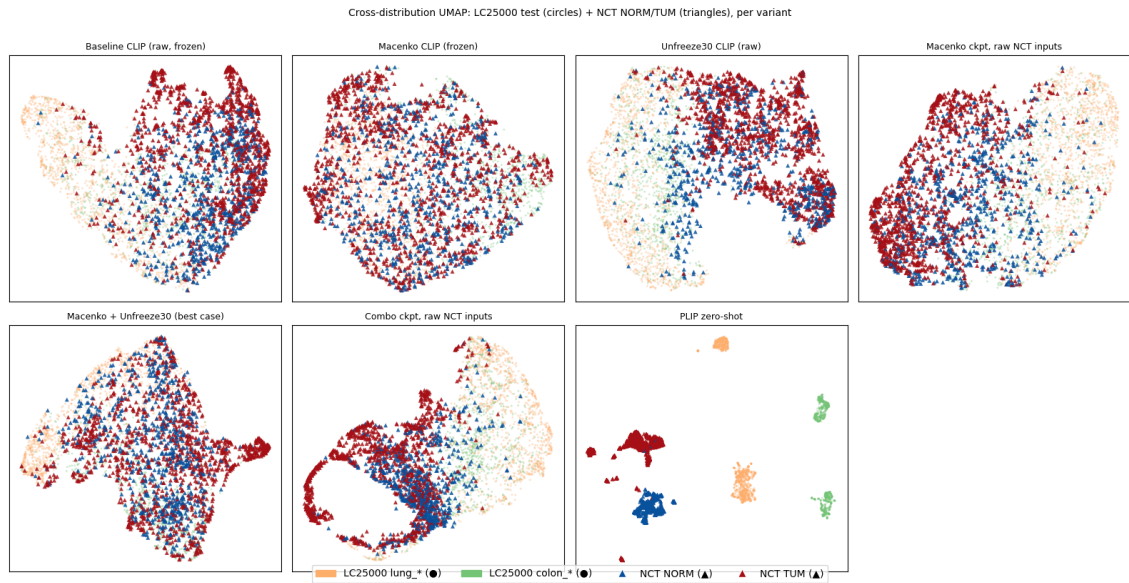


Figure 4.16 Cross-distribution UMAP per variant: LC25000 test set (circles) and NCT NORM + TUM (triangles) projected into a *single* UMAP fit per variant. Colours: LC25000 lung classes (orange), LC25000 colon classes (green), NCT NORM (dark blue), NCT TUM (dark red). Across the four working LC25000-trained variants (baseline, Unfreeze30, and the two raw-NCT disambiguation rows) the lung and colon LC25000 regions separate cleanly, and the NCT triangles — both NORM and TUM — land on the colon side, not in lung territory. *Organ-level structure transfers*: the model carries enough of LC25000’s lung-vs-colon split that NCT colon patches reach the right organ region. What does not transfer is the *within-colon* benign-vs-malignant discrimination: NORM and TUM triangles interleave in roughly the same region as the LC25000 green cluster, so the model decides “colon” but cannot pick which colon class. The two Macenko-with-Macenko variants (top-second, bottom-first) are the broken double-normalization runs and should be read with that caveat. PLIP zero-shot (bottom row, third panel) is the only variant whose UMAP cleanly separates both organ and within-organ structure into distinct clusters..

4.9 Image encoder family

- Same text encoder/prompt/split/loss/metrics; image encoder swapped.
- DINOv2 and/or OpenAI CLIP ViT-B/16; supervised vs self-supervised pretraining.
- Deferred: not run in this project.

4.10 Discussion

- Synthesis across matrix rows as they land.
- Mapping back to RQ1–RQ7 with numbers.
- Frozen general-purpose ResNet50 features cap at ~ 0.64 macro-F1 on LC25000 (measured by the plain CE classifier, §4.4). The CLIP baseline reaches 0.6273 on the same features — essentially saturating the same ceiling. Frozen-features transfer of general-purpose backbones is therefore the limiting regime, not the CLIP architecture itself.
- Pathology-domain pretraining (PLIP zero-shot) clears that ceiling by $+0.07$ macro-F1, mostly via better separation of benign vs malignant tissues. Within the frozen-encoder regime, domain pretraining is one of two levers that cross ~ 0.64 on frozen general-purpose features.
- Stain normalization is the largest input-side lever found in this study, but the gain is strongly method-dependent: Macenko clears the frozen-features ceiling by $+0.117$ macro-F1, Vahadane by $+0.037$, Reinhard only by $+0.025$ on the identical pipeline (§4.6). The gap is informative: Macenko and

Vahadane share the same stain-decomposition principle and differ only in solver (closed-form SVD vs. iterative sparse NMF), but the NMF solver fails to converge cleanly on a fraction of patches. “Stain normalization helps” therefore compresses both a real shortcut-learning effect and a methodological distinction about solver stability.

- Composing the strongest input-side lever (Macenko) with the strongest feature-adaptation regime (unfreeze30) reaches **0.918 macro-F1** (§4.6.5, Table 4.10), the highest measured in this study — +0.020 over unfreeze30 alone. The composition has diminishing returns: feature adaptation captures the bulk of the gain by shifting features into the LC25000 distribution, and Macenko’s input-side shift then adds a clean but small bonus on top. This is the variant used as the LC25000-trained checkpoint in the OOD transfer evaluation (§4.8).
- Out-of-distribution transfer (§4.8, Table 4.12) inverts the in-distribution ranking. PLIP zero-shot reaches 0.661 macro-F1 on NCT-CRC-HE-7K; the best LC25000-trained variant is raw-trained Unfreeze30 at 0.331; the Macenko-trained checkpoints fail catastrophically (0.000–0.20). The recipe that maximises in-distribution accuracy maximises domain specialisation. Within the LC25000-trained family the OOD ranking is the *opposite* of the in-distribution ranking. The cross-distribution UMAP (Figure 4.16) sharpens this further: LC25000 training does carry over the lung-vs-colon organ split, and NCT colon patches do land on the colon side of the embedding space — but the within-colon NORM-vs-TUM discrimination does not transfer. Only PLIP produces representations that visibly separate both organ and tissue-state on the external dataset (Figures 4.15, 4.16). The headline thesis-level finding is that single-institution fine-tuning of general-purpose CLIP cannot substitute for pathology pretraining when transfer requires within-organ discrimination.
- Feature-level adaptation is the dominant lever measured. Partial unfreezing of the last 30 ResNet50 layers reaches 0.898 macro-F1 with CLIP and 0.872 with plain CE. Both unlock $\sim +0.25$ macro-F1 over the frozen-features ceiling — an order of magnitude larger than any architectural or text-side change in the frozen regime.
- The CLIP architecture contributes a small, consistent +0.026 macro-F1 over plain CE under feature adaptation but ~ 0 under frozen features. Its main role is enabling capabilities (open-vocabulary classification, retrieval) rather than raw accuracy on this closed 5-class task.
- Residual confusions concentrate on lung_aca, the hardest discrimination across every variant. Even after partial unfreezing ($F_1 = 0.85$) it remains the weakest class.
- Scope boundaries: no patient-level grouping, no clinical eval, no head-to-head vs CONCH / Biomed-CLIP / MUSK, no end-to-end fine-tuning to anchor the dataset’s true ceiling.

Table 4.13 Summary of all experiments..

Experiment	Accuracy	Macro F1	Δ vs baseline
Baseline	0.6356	0.6273	—
PLIP zero-shot	0.6980	0.7004	+0.073
OpenAI CLIP zero-shot	—	—	—
Plain image classifier (frozen)	0.6440	0.6369	+0.010
Plain image classifier (unfreeze30)	0.8728	0.8723	+0.245
Partial fine-tuning (CLIP, unfreeze30)	0.8984	0.8981	+0.271
Stain normalization (Macenko, frozen)	0.7384	0.7390	+0.117
Macenko + Unfreeze30 (best case)	0.9180	0.9180	+0.291
Biomedical text encoder (best)	—	—	—
External colon (best LC25000-trained: Unfreeze30)	0.3865	0.3309	—
External colon (PLIP zero-shot)	0.5572	0.6610	—
DINOv2	—	—	—

5 Conclusions and Future Work

5.1 Conclusions

- Baseline result (RQ1): 63.6% / 62.7% macro-F1, worst class lung adenocarcinoma.
- Pathology pretraining (RQ2): PLIP +7.3 macro-F1, mainly benign vs malignant.
- Feature adaptation (RQ3): partial unfreezing (last 30 of 178 layers) +27.1 macro-F1, broad-based across classes.
- CLIP-paradigm value (RQ4): CLIP and plain CE tie at the frozen-features ceiling (~ 0.63); CLIP wins +2.6 macro-F1 over CE under feature adaptation (unfreeze30), broad-based across all classes. The paradigm's primary value remains in capabilities (open-vocabulary, retrieval), not raw accuracy.
- Stain normalization (RQ5): the gain is strongly method-specific. Macenko yields +11.7 macro-F1 ($0.627 \rightarrow 0.739$) at the frozen-backbone budget, larger than the PLIP zero-shot gain; Vahadane yields +3.7 and Reinhard +2.5 on the identical pipeline. Composing the best stain method (Macenko) with the best feature-adaptation regime (unfreeze30) reaches **0.918 macro-F1**, the highest measured in this study — +0.020 over unfreeze30 alone, so the levers compose with diminishing returns rather than additively.
- Biomedical text encoder (RQ6): *TBD*.
- OOD transfer (RQ7): the picture is two-level. The best in-distribution model (Macenko + Unfreeze30, 0.918) drops to 0.000 macro-F1 when the same Macenko transform is applied at OOD time — NCT is already colour-normalised against its own reference and the double-shift is fatal. With the OOD-time Macenko stripped, the same checkpoint reaches 0.201. The raw-trained Unfreeze30 is the best LC25000-trained transfer at 0.331; **PLIP zero-shot beats every LC25000-trained variant at 0.661 macro-F1**. The cross-distribution UMAP (Figure 4.16) reveals that LC25000 fine-tuning *does* transfer organ-level structure (NCT colon patches land in the LC25000 colon region of embedding space, not in lung), so the predictions stay within the colon classes; what does not transfer is the within-colon benign-vs-malignant discrimination, which depends on cellular cues that are more sensitive to lab/scanner/label-granularity differences. Only pathology pretraining (PLIP) produces representations that transfer at both organ *and* tissue-state level.
- Cross-cutting observations: saturation of frozen transfer, role of dataset artifacts.

5.2 Limitations

- Single-institution dataset, no patient-level grouping, no clinical validation.
- Small batch + few classes $\rightarrow$ InfoNCE false negatives, documented not mitigated.
- Frozen-backbone baseline tests transfer, not full fine-tuning.
- No human-expert evaluation.
- No head-to-head against CONCH / BiomedCLIP / MUSK (compute scope).

5.3 Future Work

Near-term:

- Additional external datasets beyond colon-Kaggle.
- SupCon or class-balanced sampling for false-negative regime.
- Patient-level grouping if metadata recoverable.

Not run in this project (low expected Δ under frozen text encoders + short prompts):

- Inference-time prompt ensembling.
- Dynamic prompt training.

Medium-term:

- Larger effective batch (grad accumulation or bigger GPU).
- Prompt-tuning methods (CoOp, CoCoOp).

Longer-term:

- Architecture-controlled comparison against CONCH / BiomedCLIP / MUSK / PathChat.
- Multi-institution training and evaluation.
- Returning to segmentation with pathology FM backbones.
- CLIP-with-short-labels applied to other low-label scientific domains.

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Bibliography

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